Piero Volpato

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PROFESSIONAL SUMMARY

Full-Stack Engineer with professional experience building web applications and APIs in C# (.NET), Python and React. Currently pursuing a **Master's degree (MSc) in Artificial Intelligence & Cybersecurity** at the University of Udine and the University of Klagenfurt, building expertise in generative AI and secure AI systems. Hands-on experience in RAG systems, local LLMs and LLM evaluation, shown in an open-source assistant with measured design choices.

FEATURED PROJECT

Grounded RAG — Local, evaluated RAG assistant over security documentation

github.com/pierobv/grounded-rag | Python, Ollama, Chroma, BM25, cross-encoder reranking, Pydantic, pytest

- Built a fully local question-answering assistant over 137 OWASP security documents that cites the source section for every claim and answers "I don't know" when the documents lack the answer. Chunking, hybrid search with reciprocal rank fusion, reranking and metrics written from scratch, without LangChain or LlamaIndex.
- Designed a 101-question evaluation set (multilingual, multi-section and out-of-corpus questions) with bootstrap confidence intervals, and ran 19 experiments to choose each component of the pipeline.
- Raised retrieval recall@5 from 83% to 95%, and from 53% to 93% on Portuguese questions, with heading-aware chunking and a multilingual embedding and reranker pair.
- Enforced citations in code with Pydantic validation (0% citations to non-existent sources) and checked the LLM-as-judge against hand-labeled verdicts (29/30 agreement). CI runs ruff, an offline test suite and Semgrep.

PROFESSIONAL EXPERIENCE

Fullstack Developer — Servizi CGN, Pordenone, Italy

February 2024 – June 2024

- Developed backend functionalities using C# (.NET) and optimized SQL queries for performance and reliability.
- Built dynamic, responsive frontend interfaces with JavaScript (React, Node.js).
- Collaborated within Agile teams to define requirements and deliver production-quality solutions.

Backend Developer (Internship) — Bidweb Security IT, Recife, Brazil

July 2022 – December 2022

- Built Python automation solutions to streamline internal processes.
- Supported the cybersecurity team in identifying and resolving vulnerabilities and implementing data-protection practices.

EDUCATION

MSc in Artificial Intelligence & Cybersecurity — University of Udine & University of Klagenfurt

September 2025 – Present (expected graduation: 2027)

Relevant coursework: Generative AI, Video Game Programming (Univ. of Udine); Social, Ethical & Legal Aspects of AI and Cybersecurity (Univ. of Klagenfurt).

Bachelor's Degree in Computer Science — UniFBV, Recife, Brazil

January 2019 – December 2022

SKILLS & LANGUAGES

Backend & Frontend: C# (.NET), Python, Node.js, RESTful APIs, React

AI / Machine Learning: RAG (hybrid search, reranking), LLM Evaluation, Ollama, Chroma, Prompt Engineering, Deep Learning, Local Generative Models (Text-to-Image/Video), AI Workflow Automation

Data & Database: SQL, Data Extraction & Cleansing (Python), Analytical Modeling

Game Development: Unity, C# Scripting

Practices: Agile Methodologies, Git/GitHub, Unit & Integration Testing, Clean Code

Languages: Portuguese (Native), English (B2), Italian (B2)

OTHER PROJECTS

- **Where2Watch** — Next.js / React / TypeScript movie-search web app with dynamic search.
- **Cinderfall** & **The Opulent Membrane** — 2D roguelike and 3D turn-based card game in Unity/C#.
- **AI & Automation Tools** — local generative AI video workflows and Python data-parsing tools.